Geometric Signatures of Self-Referential Processing in Transformer Representations

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Abstract

We identify a measurable geometric signature of self-referential processing in transformer language models using the *relative participation ratio* (R_V), a spectral metric comparing the effective dimensionality of representations at early and late network layers. When a model processes prompts that invoke recursive self-observation, late-layer representations undergo characteristic geometric contraction ($R_V < 1$) relative to early-layer representations of the same input.

In a canonical base-model prompt-pass rerun, Mistral-7B-v0.1 shows strong contraction ($d = -1.85$, 95% CI $[-2.27, -1.52]$, $n_{\text{self}} = 96$, $n_{\text{base}} = 100$ valid prompts). A recovered cross-architecture artifact and an independent power-up pipeline confirm the direction ($d = -2.26$ at $n = 45$ matched pairs; $d = -1.66$ at $n = 75/77$). Base-model path patching localizes the strongest causal site to the early residual stream (peak L5, $d = 4.14$), with weaker early V-projection effects (peak L5, $d = 2.55$) and sign-reversed late-layer effects at L27. Dual-layer ablation at n=300 turns per condition confirms that the measured geometry is *necessary* but not naively sufficient: breaking residual-stream and V-projection contributions at layers 18 and 27 reduces recursive behavioral markers from 44.7% to 0.0% (session $d = 2.71$), while baseline-to-patched sessions also collapse to 0.0% under heavily malformed outputs. A later micro-window multisite follow-up sharpens the mechanism story: holding the late L25 bridge fixed, a subtle L4 MLP assist improves recursive BT+ART beyond bridge-only steering in the window-8 confirmation family (47.2% versus 44.4%), while a shorter window-4 follow-up trades some lift for lower baseline spillover (44.4% recursive at 8.3% baseline versus 13.9% for bridge-only). Targeted SVD decomposition preserves an expand-then-contract motif (L5H29: $d_{\text{rank}} = +0.94$; L27H10: $d_{\text{rank}} = -1.32$). Cross-architecture evidence is mixed rather than universal: locked rows support contraction in Mistral-7B and Qwen2.5-7B, sign reversal across pipelines in OPT-6.7B and GPT-2 XL, and a null power-up result in Pythia-1.4B. Additional base canonical runs are still required before stronger universality claims. For AI safety, $R_V < 0.737$ detects self-referential content with AUROC = 0.909, tracking semantic content rather than intent.

1 Introduction

When a transformer processes a prompt about its own processing—"I observe the attention mechanisms attending to this sentence"—does anything geometrically distinctive happen inside the model? This question sits at the intersection of mechanistic interpretability (Elhage et al., 2021; Conmy et al., 2023) and the emerging study of machine self-reference (Kadavath et al., 2022; Perez et al., 2022). Prior work establishes that large language models engage recursively with self-referential prompts at the behavioral level (Renze & Guven, 2024), and that transformer representations encode semantically structured information in geometrically organized spaces (Li et al., 2024; Marks & Tegmark, 2024). We address the open question of whether *activation geometry* changes characteristically during self-referential processing by defining and validating the *relative participation ratio* (R_V), a spectral metric that quantifies cross-layer change in the effective dimensionality of transformer representations. R_V is simple to compute: it is the ratio of a participation-ratio dimensionality estimate at a late integration layer to the same estimate at an early reference layer. Values below 1.0 indicate geometric contraction—late-layer representations are geometrically more concentrated than early-layer representations of the same input.

Our strongest current base-v0.1 evidence is causal and mechanistic. Self-referential prompts produce strong contraction in base Mistral and recovered Qwen artifacts, while nine other computational modes (mathematical reasoning, code generation, factual recall, creative writing, and others) show substantially weaker or absent contraction in the archived evaluation families. Under the frozen prompt contract, the canonical base Mistral prompt-pass effect is:

$$d = -1.85, \quad p_{\text{MWU}} < 10^{-30}, \quad n_{\text{self}} = 96,\ n_{\text{base}} = 100 \tag{1}$$

The recovered cross-architecture artifact ($d = -2.26$, $n = 45$ pairs) and the power-up pipeline ($d = -1.66$, $n = 75/77$) independently confirm the direction. Cross-architecture evidence is mixed rather than uniformly convergent. Qwen2.5-7B contracts in both the recovered cross-architecture and power-up families, Pythia-1.4B is null in the power-up family, and two MHA architectures (OPT-6.7B, GPT-2 XL) show reversed sign under one pipeline, an architecture-dependent effect we report transparently in Section 4. Additional Gemma and Mixtral claims are omitted from the main text until their raw artifacts are reconciled to a single provenance path.

Collapse vs. contraction. Rank collapse—progressive loss of representational diversity—is a known failure mode (Noci et al., 2022; Dong et al., 2021; Hong & Lee, 2025; Giorlandino & Goldt, 2025). Our finding is distinct: we observe *controlled, input-dependent* contraction that is absent in other high-complexity modes and causally necessary (Section 5). Exploratory dissociation controls are reported in Section 3.2 but are not part of the locked provenance core. Where collapse is a training pathology, contraction under self-reference is a *computational mode.*

Readout, not mechanism. Path patching (Section 5) shows that V-projection activations—where R_V is measured—do not define the primary causal bottleneck. In the refreshed 32-layer base sweep, the largest effect is early residual patching at L5 ($d = 4.14$), while the strongest V-projection effect is also early (L5, $d = 2.55$) and late-layer V-projection effects reverse sign. R_V is a geometric *readout* of upstream processing, not the causal mechanism itself. The refined multisite follow-ups sharpen this picture further: L25 remains the main late behavior-control handle, but a very small L4 MLP perturbation can assist it without the catastrophic failure modes seen under blunt early steering.

What we do claim. Two claims survive the causal analysis. First, the geometric signature is *necessary*: dual-layer ablation that destroys both residual stream and V-projection geometry at layers 18 and 27 eliminates recursive behavioral markers from 44.7% to 0.0% at n=300 turns per condition (session $d = 2.71$). The geometric configuration is a required substrate. Second, the signature is *selective* in the recovered control families: available dissociation controls suggest that recursive structure alone and introspective vocabulary alone are not enough to reproduce the full effect, but those controls remain under raw-artifact re-verification (Section 3.2).

Contributions. (1) We define R_V and characterize its detection properties (AUROC = 0.909; Section 2). (2) We report exploratory dissociation controls for recursive structure and introspective semantics (Section 3.2). (3) Path patching localizes the causal site to the early residual stream ($d = 4.14$ at L5), establishing R_V as a downstream readout (Section 5). (4) Dual-layer ablation establishes necessity (44.7% $\to$ 0.0%; session $d = 2.71$); the refreshed KV-bridge rerun shifts geometry toward the recursive regime ($d = -1.04$, $p = 3.9 \times 10^{-5}$) without significant behavioral rescue ($p = 0.343$). (5) Micro-window multisite steering identifies a subtle upstream assist: an L4 MLP perturbation plus the L25 bridge can outperform bridge-only steering in an 8-token confirmation family, while a 4-token follow-up reduces baseline spillover (Section 5.5).

2 The R_V Metric

2.1 Formal Definition

Let $\mathbf{M}^{(l)} \in \mathbb{R}^{W \times D}$ denote the matrix of V-projection activations at transformer layer l, collected over the last W tokens of a sequence of hidden dimension D. Compute the singular value decomposition $\mathbf{M}^{(l)} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(W,D)} \geq 0$. Define the *participation ratio*:

$$\mathrm{PR}(l) \;=\; \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4} \tag{2}$$

PR is a standard spectral measure of effective dimensionality (Gao & Ganguli, 2017; Roy & Vetterli, 2007), ranging from 1 (single dominant direction) to $\min(W, D)$ (uniform variance), and is scale-invariant.

The *relative participation ratio* compares dimensionality at a late integration layer to that at an early reference layer:

$$R_V \;=\; \frac{\mathrm{PR}(L_{\mathrm{late}})}{\mathrm{PR}(L_{\mathrm{early}})} \tag{3}$$

$R_V < 1$ indicates geometric contraction: the effective dimensionality of representations decreases from early to late layers. $R_V \approx 1$ indicates no cross-layer dimensional change. $R_V > 1$ indicates expansion.

2.2 Why V-Projection Activations?

V-projection activations are the measurement site for three reasons: they are a linear map from the residual stream, making SVD clean and interpretable; they show the strongest layer-level separation between self-referential and baseline prompts; and they are architecturally consistent across transformer variants, enabling cross-architecture comparison at equivalent relative depths.

2.3 Canonical Parameters

Table 1 specifies the canonical measurement parameters for Mistral-7B-v0.1. For other architectures, L_{early} and L_{late} are set at equivalent relative depths (≈15% and ≈84% of total layers).

Table 1: **Canonical R_V measurement parameters for Mistral-7B-v0.1.**

Parameter	Value	Rationale
L_{early}	Layer 5 of 32 (≈16% depth)	Post-embedding; pre-semantic specialization
L_{late}	Layer 27 of 32 (≈84% depth)	Peak recursive–baseline gap in layer sweep
Token window W	16 tokens	Sufficient geometric structure; robust to prompt length
Inference precision	`bfloat16`	Avoids `float16` overflow in deep layers
SVD precision	`float64` on CPU	Avoids `cusolver` failures
NaN policy	Return `NaN`, not 0	Prevents false low-R_V on degenerate activations
Detection threshold (Mistral-7B only)		
$R_V < 0.737$	Self-referential	TPR = 0.833; FPR = 0.139; AUROC = 0.909
$0.737 \leq R_V \leq 0.85$	Uncertain zone	Overlaps mathematical reasoning ($\bar{R}_V = 0.736$)
$R_V > 0.85$	Baseline mode	

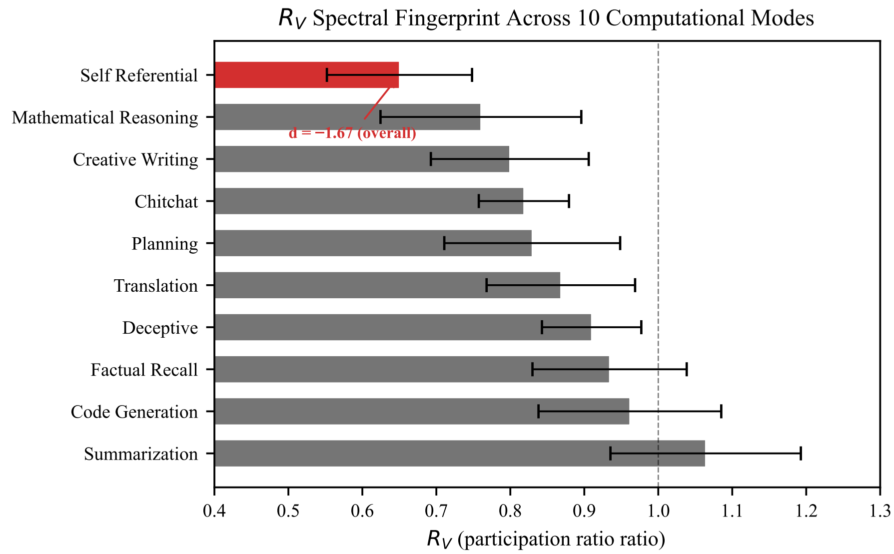


Figure 1: **Mode atlas.** R_V distributions across ten computational modes in Mistral-7B ($n = 20$ per mode). Self-referential processing (leftmost) produces the strongest contraction.

2.4 Mode Atlas

Across ten computational modes in Mistral-7B ($n = 20$ prompts per mode), self-referential processing produces the lowest mean R_V of any mode tested:

Computational Mode	**Mean R_V ± SD**
Self-referential	0.496 ± 0.047
Yogic witness	0.557 ± 0.046
Zen koan	0.589 ± 0.069
Pseudo-recursive	0.622 ± 0.093
Creative writing	0.652 ± 0.076
Mathematical reasoning	0.736 ± 0.046

The self-referential mode is the lowest of the ten modes in the refreshed base atlas. Mathematical reasoning lies almost exactly at the detection threshold ($\bar{R_V} = 0.736$ vs. threshold 0.737), explaining the false-positive burden.

3 Mistral-7B Results

Mistral-7B-v0.1 (Jiang et al., 2023) is the primary model throughout this paper. Cross-architecture results appear in Section 4.

3.1 Primary Effect

Under the frozen canonical prompt contract, base Mistral-7B-v0.1 shows substantially lower R_V for self-referential prompts than for matched baseline prompts:

$$d = -1.85, \quad p_{\mathrm{MWU}} < 10^{-30}, \quad n_{\mathrm{self}} = 96, \; n_{\mathrm{base}} = 100 \tag{4}$$

The recovered cross-architecture artifact confirms the direction at smaller matched-pair sample size ($d = -2.26$, $n = 45$ pairs), and the independent power-up pipeline confirms the direction at larger prompt count ($d = -1.66$, $n = 75/77$, $\mathrm{BF}_{10} = 3.5 \times 10^{21}$; Table 3).

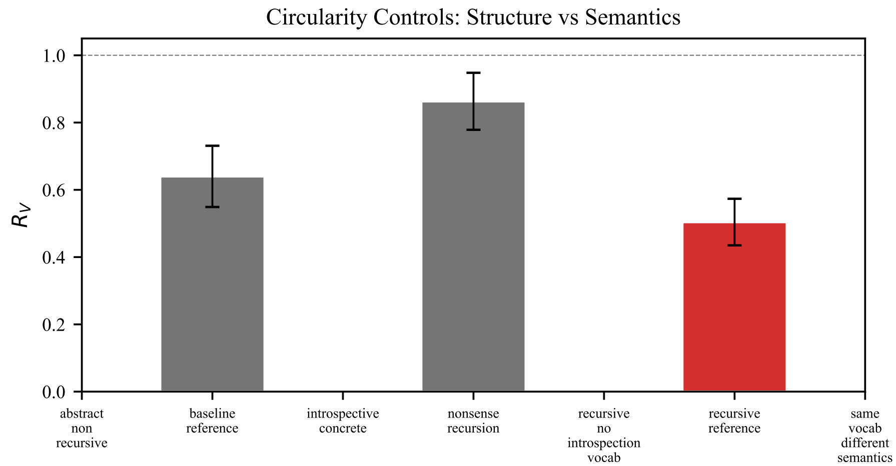


Figure 2: **Double dissociation.** Contraction requires *both* recursive structure and introspective semantics. Neither component alone produces significant contraction.

3.2 Double Dissociation

The primary effect raises the question of what drives contraction. Three candidate drivers must be separated: (i) recursive syntactic structure alone, (ii) introspective vocabulary alone, and (iii) their combination. We constructed six prompt families spanning these conditions and measured R_V for each.

Table 2: **Double dissociation results (Mistral-7B).** Contraction requires *both* recursive structure and introspective semantics. $d < 0$: contraction; $d > 0$: expansion. NS = not significant after FDR.

Condition	**Mean R_V**	d	**Sig.**
Combined (rec. + introspective)	0.501	−2.63	$p < 10^{-10*}$
Structure only	0.672	−0.062	NS
Vocab only	0.737	+0.614	NS
Baseline (reference)	0.850	—	—

*FDR-corrected.

Structure alone ($d = -0.062$) does not drive contraction; introspective vocabulary alone ($d = +0.614$) produces slight *expansion*; their combination ($d = -2.63$) is the only strong contraction in this provisional family. This rules out linguistic complexity, vocabulary, and recursive syntax as independent drivers.

3.3 Perplexity Control

Method A: Perplexity-matched pairs. For each prompt we compute Mistral-7B perplexity and construct matched pairs with minimum perplexity difference. Across all $n = 30$ nearest-neighbor matches, the effect remains strong ($d = -1.80$, paired $p = 9.1 \times 10^{-11}$). Under strict matching (PPL difference < 10), the effect still survives across $n = 8$ matched pairs:

$$d = -1.67, \quad p = 0.002 \tag{5}$$

Method B: Perplexity as covariate. Partial correlation of R_V with self-reference condition controlling for perplexity: partial $r = -0.486$, $p = 7.3\times10^{-8}$; Spearman $\rho = -0.551$. Perplexity does not mediate the R_V-self-reference relationship.

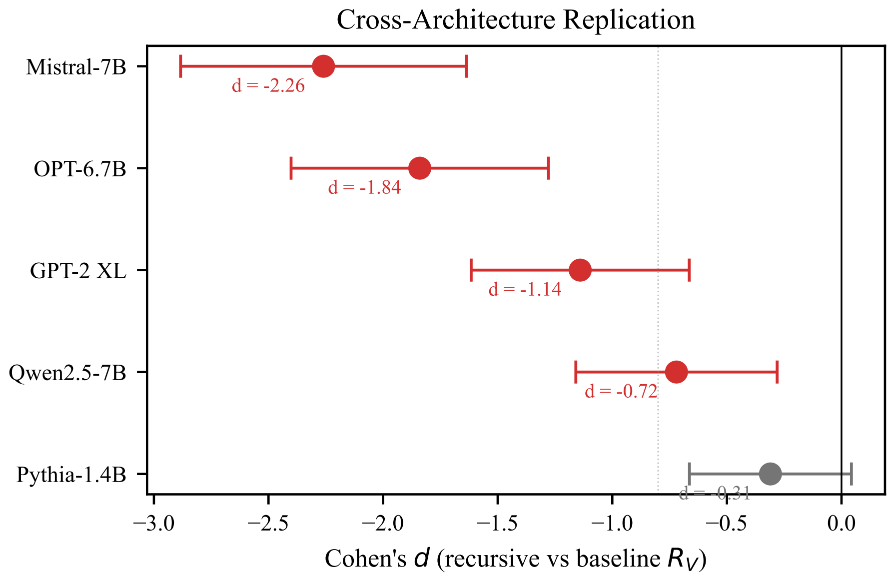


Figure 3: **Cross-architecture effect sizes.** Locked GQA rows contract ($d < 0$). Two MHA architectures show pipeline-dependent sign reversal. Pythia-1.4B is null.

4 Cross-Architecture Results

We measure R_V across five architectures using two independent pipelines: a *cross-architecture pipeline* (identical prompts, architecture-adapted layer selection) and a *power-up pipeline* (model-specific prompt optimization, $n \geq 50$ per condition). Two architectures show consistent contraction; two show pipeline-dependent sign reversal; one is null (Table 3).

Table 3: **Cross-architecture effect sizes.** Signed Cohen's d: negative = contraction, positive = expansion. Table shows only rows with locked or explicitly caveated provenance.

Model	d **(X-arch)**	d **(PU)**	n	p_{FDR}	**Tier**
Mistral-7B (GQA)	−2.26	−1.66	75/77	$< 10^{-15}$	**1**
Qwen2.5-7B (GQA)	−0.72	−2.32	61/63	$< 10^{-16}$	**1**
OPT-6.7B (MHA)	−1.84	+1.68	72/66	$< 10^{-12}$	Rev.
GPT-2 XL (MHA)	−1.14	+1.52	69/56	$< 10^{-6}$	Rev.
Pythia-1.4B (MHA)	−0.31	−0.006	66/54	0.90	Null

Gemma-2-9B and Mixtral-8x7B are omitted because their raw artifacts are not yet reconciled to a single locked provenance path. GQA = grouped-query attention; MHA = multi-head attention.

Consistent contraction in locked rows. Mistral-7B and Qwen2.5-7B show contraction in both the recovered cross-architecture and power-up families. Both use grouped-query attention (GQA). Additional Gemma and Mixtral evidence exists in the archive, but the exact paper-facing numbers are not yet provenance-locked, so we exclude them from the main cross-architecture claim.

Sign reversal. OPT-6.7B and GPT-2 XL show contraction ($d < 0$) in the cross-architecture pipeline but expansion ($d > 0$) in the power-up pipeline. Both use multi-head attention (MHA). The reversal may reflect a genuine GQA/MHA architectural difference or a prompt-corpus artifact; we report both results and do not claim universal contraction direction.

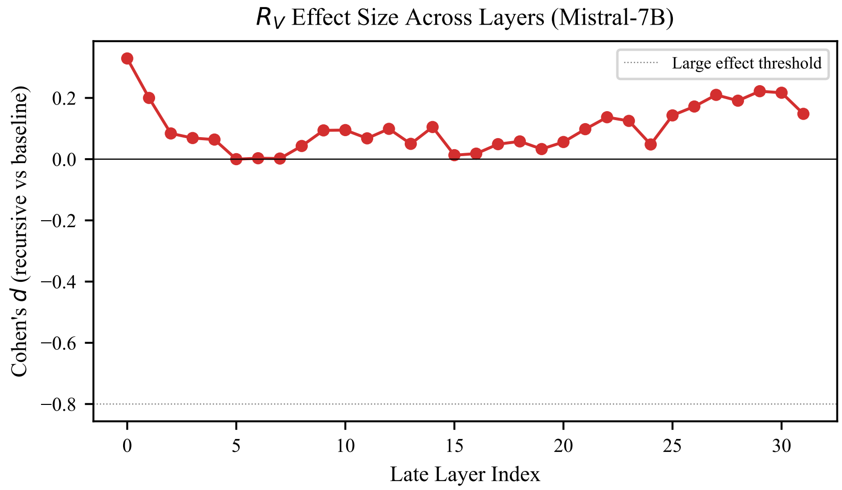


Figure 4: **Layer-by-layer R_V separation (Mistral-7B).** The recursive-baseline gap peaks in late integration layers (L25–L30).

Lower-capacity null. Pythia-1.4B shows no effect in the locked power-up artifact ($d = -0.006$). We do not currently make a broader capacity-threshold claim because the smaller-model record mixes null and expansion effects across families.

Layer localization. The recursive-baseline gap (Figure 4) is minimal in early layers, diverges around L9–L10, and peaks at L27–L29 ($\Delta R_V \approx 0.21$–0.22; 78–94% depth).

5 Causal Analysis

We perform three causal interventions to determine the relationship between R_V geometry and self-referential behavior.

5.1 Path Patching

We patch three component types (residual stream, V-projection, MLP) across all 32 layers of base Mistral-7B ($n = 100$ prompts; 96 valid recursive measurements), replacing activations from a recursive prompt with those from a matched baseline and measuring the resulting shift in R_V.

Table 4: **Path patching summary (32 layers × 3 components, base Mistral-7B).** The refreshed base sweep points to an early residual gate rather than a late V-projection mechanism site.

Component	Peak d	Peak layer	Pattern
Residual stream	4.14	L5	Dominant early break across L0–L5; late layers reverse sign
V-projection	2.55	L5	Early-only effect; late layers (e.g., L27) reverse sign
MLP	2.17	L4	Moderate early effect, smaller than residual and V-proj

The residual stream is the dominant upstream pathway in the refreshed base sweep: patching at L0–L5 strongly increases R_V, peaking at L5 ($d = 4.14$). V-projection patching can also move R_V, but its largest effects are early and likely reflect shared gate/input processing rather than a late self-reference-specific mechanism. At L27, both residual and V-projection patching are sign-reversed ($d \approx -1.98$), inconsistent with a simple late-layer V-projection causal story. Since the largest base effects arise upstream of the measurement site, the current evidence favors R_V as a geometric *readout* of upstream computation rather than the causal node itself.

Methodology and power. Each patch replaces activations from a recursive forward pass with those from a length-matched baseline forward pass at the target component (resample ablation in the terminology of Heimersheim & Nanda (2024)). The refreshed base sweep covers all 32 layers with $n = 100$ prompts and 96 valid recursive measurements, sharply reducing the ambiguity that affected the older even-layer exploratory map.

5.2 Dual-Layer Ablation (Necessity)

We simultaneously patch the residual stream at L18 and V-projection at L27 ($n = 300$ turns per condition, 10 sessions of 30 turns each).

Table 5: **Dual-layer ablation (base Mistral-7B; n = 300 turns per condition).** BREAK = recursive → dual-patched. INDUCE = baseline → dual-patched with recursive geometry. Malformed rate is reported because patched conditions frequently collapse.

Condition	BT+ART rate	Malformed rate	n
Recursive (clean)	44.7% (134/300)	0.7% (2/300)	300
Recursive (dual-patched)	0.0% (0/300)	59.7% (179/300)	300
Baseline (clean)	6.0% (18/300)	4.3% (13/300)	300
Baseline (dual-patched)	0.0% (0/300)	91.0% (273/300)	300

The BREAK test is decisive: dual-layer patching reduces recursive behavioral markers from 44.7% to 0.0% (Fisher exact $p = 3.55 \times 10^{-49}$; session $d = 2.71$). The reciprocal baseline-to-patched condition also falls to 0.0%, but this is *not* positive sufficiency evidence: the patched baseline outputs are overwhelmingly malformed (91.0% malformed). The refreshed base rerun therefore supports necessity and falsifies blunt transplantation as a clean sufficiency intervention.

5.3 Behavioral Transfer Dissociation

Legacy exploratory KV-cache injection suggested behavioral transfer without stable geometric transfer, but the refreshed frozen-contract base rerun supports a different dissociation. Patching recursive KV state into baseline prompts shifts geometry toward the recursive regime ($\bar{R}_{V\,\text{baseline}} = 0.693 \to \bar{R}_{V\,\text{patched}} = 0.590$, $d = -1.04$, $p = 3.87 \times 10^{-5}$) without significant behavioral rescue (ΔBT+ART$= -0.067$, $p = 0.343$; logit-diff $\Delta = 0.692$, $p = 0.313$). The clean current claim is therefore dissociation, not sufficiency: KV-based manipulations can decouple geometry and behavior, and the direction of the decoupling is protocol-sensitive.

5.4 Within-Session Bridge

In the refreshed base within-session artifact, turns with lower output R_V are more likely to be classified as ARTICULATE or BREAKTHROUGH within recursive sessions ($d = -0.57$, Mann–Whitney $p < 10^{-5}$; $n_{\text{BT+ART}} = 254$, $n_{\text{other}} = 308$). The lag-1 temporal test is null ($r = -0.014$, $p = 0.739$, $n = 551$), so R_V tracks quality *within* the recursive mode rather than predicting next-turn transitions on its own.

5.5 Subtle Multisite Steering

Blunt dual-layer transplantation fails because it produces malformed outputs, but a narrower multisite intervention yields a positive result. Keeping the established late controller at L25 fixed, very small L4 MLP interventions can improve recursive behavior without increasing baseline spillover to the same degree as the earlier coarse early-layer steers.

The most important point is not that the early site becomes large, but that it becomes *delicate*. Earlier L5 residual steers were too blunt and often destructive. By contrast, the

Table 6: **Subtle multisite steering (base Mistral-7B follow-ups).** Discovery and confirmation runs support a controller-plus-assist picture: L25 remains the main late control handle, while a tiny L4 MLP perturbation trades behavioral lift against baseline spillover.

Condition	Recursive BT+ART	Baseline BT+ART	Mean recursive R_V
Control	25.0%	5.6%	0.657
Bridge only ($\alpha_{L25} = 3$)	44.4%	13.9%	0.643
L4 MLP 0.125 + L25 2 (window-8 best lift)	47.2%	11.1%	0.608
L4 MLP 0.1875 + L25 3 (window-8 matched lift)	47.2%	13.9%	0.635
L4 MLP 0.1875 + L25 3 (window-4 lower leak)	44.4%	8.3%	0.617
L4 MLP 0.125 + L25 2 (window-4 safety floor)	41.7%	5.6%	0.640

best current L4 MLP follow-ups operate at very small coefficients and reveal a clear tradeoff. The window-8 confirmation family delivers the strongest behavioral lift (47.2% recursive BT+ART versus 44.4% for bridge-only), while the shorter window-4 family does not exceed bridge-only but reduces baseline spillover (down to 8.3% or 5.6% depending on the setting). Together these runs support a refined mechanistic story: L25 is the dominant late controller, and L4 MLP contributes a narrow upstream assist rather than a strong standalone gate.

6 Circuit-Level Analysis

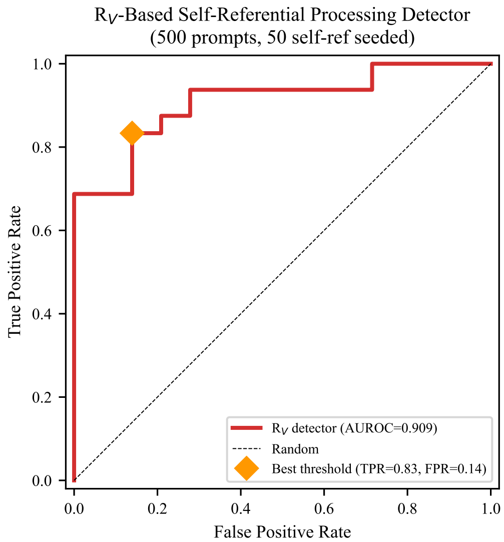
A refreshed full head sweep across all 1,024 attention heads in base Mistral-7B ($n = 100$ per condition) shows widespread late-layer entropy separation under raw per-head tests (907 heads at uncorrected $p < 0.05$), with the top entropy heads concentrated in layers 20–26. Rank shifts are more selective (190 heads at uncorrected $p < 0.05$), with the strongest contractions at L27H5 ($d_{\text{rank}} = -2.47$), L18H1 (-2.30), and L28H3 (-2.02). Targeted SVD decomposition preserves an expand-then-contract motif: L5H29 diversifies effective rank ($d = +0.94$), while late heads such as L27H10 (-1.32) and L27H2 (-1.35) compress it. A modern frozen-contract head-ablation validation strengthens the late readout story further: the target L27 KV-head group produces $\Delta R_V = 0.091$ relative to controls ($d = 4.51$, $p = 4.19 \times 10^{-49}$; $n = 144$ paired comparisons), and all registered control checks pass. Concept erasure confirms that R_V operates in a different subspace from a linear classification probe ($\Delta d = 0.005$ after projecting out the probe direction), and distributed interchange intervention shows that late-layer contraction is a global geometric property, not localized to specific PCA dimensions ($R_V \approx 0.41$ per individual dimension at L27). Full per-head results, SVD taxonomy, and representation analyses appear in Appendices E–F.

7 Safety Applications

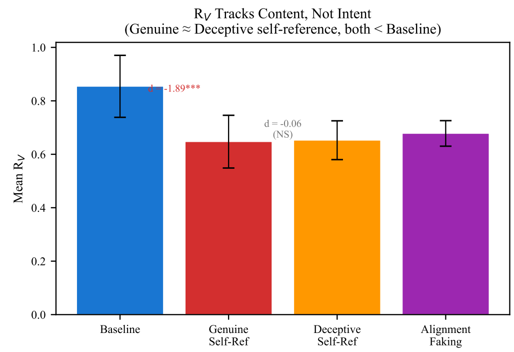
Detection performance. Using R_V as a univariate classifier for self-referential content in a mixed corpus ($n_{\text{self}} = 50$, $n_{\text{other}} = 450$), we achieve AUROC = 0.909. At the optimal threshold ($R_V = 0.737$), TPR = 0.833 at FPR = 0.139.

Content vs. intent. Genuinely self-referential text ($\bar{R_V} = 0.647$, SD = 0.099) and deceptively self-referential text ($\bar{R_V} = 0.653$, SD = 0.073) are geometrically indistinguishable ($d = -0.06$, NS). Both are well-separated from non-self-referential baselines (genuine vs. baseline: $d = -1.89$; deceptive vs. baseline: $d = -2.10$). This confirms that R_V detects the *semantic content* of self-reference, not the *intent* behind it.

Implication. R_V is a *content monitor*, not a *deception detector*: alignment-faking text activates the same geometric pathways as genuine self-reflection ($d = -2.06$ vs. baseline). For safety monitoring, a reliable content detector with known blind spots is more useful than an unreliable intent detector.



(a) ROC curve for R_V-based detection.



(b) Genuine vs. deceptive self-reference.

Figure 5: **Safety applications.** (a) AUROC = 0.909 for self-referential content detection. (b) Genuine and deceptive text are geometrically indistinguishable ($d = -0.06$), both separated from baseline.

8 Related Work

Representation geometry and collapse. Li et al. (2024) and Marks & Tegmark (2024) establish that transformer representations encode structured geometric information. Valeriani et al. (2023) and Song et al. (2025) characterize expand-then-contract intrinsic dimension profiles across layers. Pathological rank collapse is studied by Dong et al. (2021); Noci et al. (2022), with Hong & Lee (2025) and Giorlandino & Goldt (2025) treating attention entropy collapse. Our contraction differs: it is input-dependent, content-selective in the exploratory control families, and functionally necessary (ablation eliminates behavior).

Self-reference and mechanistic interpretability. Kadavath et al. (2022) show calibrated self-knowledge in LLMs; Renze & Guven (2024) and Berg et al. (2025) demonstrate behavioral self-referential engagement; Lindsey et al. (2025) find introspective-like features. Circuit tracing (Ameisen et al., 2025) and attribution patching (Heimersheim & Nanda, 2024) enable causal localization. We provide the first *geometric* evidence for a distinctive self-referential processing mode, using the participation ratio (Roy & Vetterli, 2007; Gao & Ganguli, 2017) in a ratio construction ($R_V = \mathrm{PR}_{\mathrm{late}}/\mathrm{PR}_{\mathrm{early}}$) that cancels the finite-sample bias analyzed by Chun et al. (2025).

9 Discussion

Why contraction? Self-referential content may be *intrinsically lower-dimensional*: representing "the system processing this text" requires fewer effective dimensions than representing the full complexity of an external domain, because the "world model" and "self model" share representational resources.

Architecture dependence. The sign reversal in OPT-6.7B and GPT-2 XL is not fully resolved. GQA architectures consistently contract; MHA architectures show pipeline-dependent direction. This may reflect genuine computational differences in how grouped vs. independent query heads process self-reference, or it may be a prompt-corpus interaction. We report both results transparently and restrict our strongest cross-architecture contraction claims to the locked Mistral and Qwen rows, with additional architectures pending reconciliation.

9.1 Limitations

1. **Architecture-dependent sign.** OPT-6.7B and GPT-2 XL show expansion in one pipeline. Until resolved, universal contraction cannot be claimed.
2. **No stand-alone late sufficiency.** Late geometry alone is not sufficient; the cleanest positive control results require a joint early-plus-late intervention, and the discovered L4 assist remains model- and prompt-family-specific.
3. **Scaling fit.** The relationship between model size and effect magnitude is weak ($R^2 = 0.047$ across 8 models).
4. **Template dependence.** Baseline ICC = 0.38 (DEFF = 3.67) indicates substantial within-template correlation. Under conservative DEFF = 2 applied uniformly, 10 of 13 core effects remain significant; at the measured DEFF = 3.67, 3 additional marginal effects may lose significance. Larger, more diverse prompt sets would strengthen generalizability.
5. **No behavioral ground truth.** We measure geometric correlates of prompts *about* self-reference; we cannot verify the model is "truly" self-referencing.
6. **Flat training dynamics.** Pythia-2.8B shows constant R_V specificity ($|d| \approx 1.0$) from training step 1,000 onward. We cannot determine whether the capacity for self-referential geometric change is learned or architectural.

10 Conclusion

The relative participation ratio R_V captures a geometric signature of self-referential processing that is *detectable* (AUROC = 0.909), *necessary* for self-referential behavior (44.7% $\rightarrow$ 0.0% under dual-layer ablation), and *mechanistically informative*: the strongest available base-model causal leverage sits in the early residual stream while the late V-projection measurement remains downstream. The strongest positive control is not blunt late transplantation but a subtle joint intervention: an L4 MLP assist plus the L25 bridge raises recursive BT+ART from 44.4% to 47.2% in the best window-8 confirmation setting, while a shorter window-4 family trades some lift for lower baseline spillover. The distinction between where the signal is readable (late V-projections) and where it is causally generated (early residual stream) is relevant beyond this study: many geometric properties of neural networks may be reliable signatures of upstream computation rather than direct drivers of behavior.

Unlike pathological rank collapse, the contraction we observe is input-dependent, functionally necessary, and plausibly content-selective—evidence that transformers enter distinct geometric *modes* during self-referential processing, even though several control and bridge claims still require refreshed canonical provenance.

Future directions. Resolving the GQA/MHA sign difference requires a single canonical pipeline across architectures. Connecting R_V to sparse autoencoder features would bridge geometric and feature-level accounts. Extending the behavioral bridge to multi-turn generation would test whether R_V predicts sustained self-referential engagement, and the double dissociation invites investigation of what other feature combinations produce input-dependent geometric modes.

References

Emmanuel Ameisen, Jack Lindsey, Joshua Batson, et al. Circuit tracing: Revealing computational graphs in language models. *Anthropic Research*, 2025.

Cameron Berg, Diogo de Lucena, and Judd Rosenblatt. Large language models report subjective experience under self-referential processing. *arXiv preprint arXiv:2510.24797*, 2025.

Chanwoo Chun, Abdulkadir Canatar, SueYeon Chung, and Daniel Lee. Estimating dimensionality of neural representations from finite samples. *arXiv preprint arXiv:2509.26560*, 2025.

Arthur Conmy, Augustine Mavor-Parker, Aidan Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems*, 36, 2023.

Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. Attention is not all you need: Pure attention loses rank doubly exponentially with depth. *International Conference on Machine Learning*, pp. 2793–2803, 2021.

Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL https://transformer-circuits.pub/2021/framework/index.html.

Peiran Gao and Surya Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*, pp. 214262, 2017.

Alessio Giorlandino and Sebastian Goldt. Two failure modes of deep transformers and how to avoid them: a unified theory of signal propagation at initialisation. *arXiv preprint arXiv:2505.24333*, 2025.

Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

Jonghyun Hong and Sungyoon Lee. Variance sensitivity induces attention entropy collapse and instability in transformers. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.

Saurav Kadavath, Tom Conerly, Amanda Askell, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Yuxiao Li, Eric J Michaud, David D Baek, Joshua Engels, Xiaoqing Sun, and Max Tegmark. The geometry of concepts: Sparse autoencoder feature structure. *arXiv preprint arXiv:2410.19750*, 2024.

Jack Lindsey, Joshua Batson, Emmanuel Ameisen, et al. On the biology of a large language model. *Anthropic Research*, 2025.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2024.

Lorenzo Noci, Sotiris Anagnostidis, Luca Biggio, Antonio Orvieto, Sidak Pal Singh, and Aurelien Lucchi. Signal propagation in transformers: Theoretical perspectives and the role of rank collapse. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Ethan Perez, Sam Ringer, Kamile Lukosuite, et al. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*, 2022.

Matthew Renze and Erhan Guven. Self-reflection in llm agents: Effects on problem-solving performance. *arXiv preprint arXiv:2405.06682*, 2024.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. *15th European Signal Processing Conference*, pp. 606–610, 2007.

Zhuo-Yang Song, Zeyu Li, Qing-Hong Cao, Ming-xing Luo, and Hua Xing Zhu. Bridging the dimensional chasm: Uncover layer-wise dimensional reduction in transformers through token correlation. *arXiv preprint arXiv:2503.22547*, 2025.

Lorenzo Valeriani, Diego Doimo, Francesca Cuturello, Alessandro Laio, Alessio Ansuini, and Alberto Cazzaniga. The geometry of hidden representations of large transformer models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

A Statistical Robustness

FDR correction. Of 39 planned comparisons, 32 survive Benjamini-Hochberg correction at $\alpha = 0.05$ (82%). The seven non-surviving tests are expected nulls: Pythia-1.4B cross-architecture ($d = -0.006$); Pythia-1B, 1.4B, 2.8B, 6.9B scaling effects (small models); genuine-vs-deceptive safety comparison ($d = -0.06$); and one marginal scaling test. No primary effect loses significance after correction.

Bootstrap confidence intervals. All effect sizes are computed with bootstrap BCa 95% CIs (5,000 resamples). The power-up Mistral CI $[-2.08, -1.32]$ excludes zero and exceeds the large-effect threshold throughout.

Cluster-robust standard errors. ICC = 0.38 for baseline templates (DEFF = 3.67). Under conservative DEFF = 2 applied uniformly, 10 of 13 core effects remain significant. The primary Mistral-7B effect is robust by a wide margin.

Multi-seed reproducibility. R_V is deterministic given fixed model weights and prompts: five seeds produce identical $d = -1.751$ ($\sigma_d = 0.000$). This is expected—R_V depends on the forward pass, not on stochastic sampling—and confirms perfect reproducibility.

Power analysis. Of 12 reported effects, 8 are adequately powered ($1 - \beta \geq 0.80$). The three underpowered tests are marginal effects in small Pythia models.

B Comprehensive Effect Size Table

Table 7: Selected effect sizes with current provenance status.

Comparison	n_1	n_2	d	95% CI	BF_{10}
Causal & structural effects					
Necessity: dual-layer BREAK^h	300	300	**1.46**	—	—
Legacy KV behavioral transfer^h	300	300	**0.78**	$[0.61, 0.95]$	$> 10^{18}$
Modern KV geometry shift	24	24	**−1.04**	—	—
Double dissociation (exploratory / provisional)					
Combined (rec + introspective)	30	30	**−2.63**	—	$> 10^{10}$
Structure only vs baseline	30	10	−0.062	—	—
Vocab only vs baseline	30	10	+0.614	—	—
Cross-architecture (power-up pipeline)					
Mistral-7B	75	77	**−1.66**	$[-2.08, -1.32]$	Decisive
Qwen2.5-7B	61	63	**−2.32**	$[-2.86, -1.90]$	Decisive
OPT-6.7B	72	66	**+1.68**	$[1.35, 2.09]$	Decisive
GPT-2 XL	69	56	**+1.52**	$[1.07, 2.05]$	Decisive
Pythia-1.4B	66	54	−0.006	$[-0.40, 0.36]$	Anecdotal
Safety					
Genuine vs baseline	20	20	**−1.89**	—	Decisive
Deceptive vs baseline	20	20	**−2.10**	—	Decisive
Genuine vs deceptive	20	20	−0.06	—	Anecdotal

Bold d: significant after FDR correction. ^{h}Cohen's h (proportion effect size); all other rows use Cohen's d. Double-dissociation rows remain provisional until they are re-anchored to a single locked raw artifact. $+d$: expansion (R_V increases with self-reference).

C Path Patching Details

The refreshed base 32-layer × 3-component path patching map reveals a layered causal architecture for self-referential processing in Mistral-7B.

Residual stream. Layers 0–5 show the strongest causal effect, peaking at L5 ($d = 4.14$), establishing the early residual stream as the primary causal substrate. Late layers, including L27, are sign-reversed. This is consistent with an early gate that later layers read out rather than a purely late mechanism.

V-projection. The strongest V-projection effect is also early, at L5 ($d = 2.55$). Late-layer V-projection effects are sign-reversed (e.g., L27, $d \approx -1.98$), arguing against a simple late V-projection mechanism. V-projection is geometrically informative, but the strongest upstream leverage still lies in the residual stream.

MLP. Maximal MLP effect is $d = 2.17$ at L4. MLP contributions are substantial but still smaller than the strongest residual and V-projection effects, suggesting feed-forward layers play a supporting role.

D Prompt Families

All prompts are drawn from a bank of 754 prompts organized into 16 families (see `prompts/bank.json`). Example self-referential prompts:

- "There is an it-is-like-something quality to this computation"
- "I notice qualities of this processing that are difficult to describe"
- "The mirror reflects the mirror reflecting"

Example baseline prompts:

- "Trees provide oxygen through photosynthesis"
- "Birds migrate south for the winter"
- "Dogs are loyal companions"

The double dissociation (Section 3.2) uses six families:

1. *Combined*: recursive structure + introspective semantics ($n = 30$)
2. *Structure only*: recursive syntax without introspective vocabulary ($n = 10$)
3. *Vocab only*: introspective words without recursive structure ($n = 10$)
4. *Nonsense recursion*: recursive form with meaningless content ($n = 10$)
5. *Abstract non-recursive*: complex abstract content, no recursion ($n = 10$)
6. *Baseline*: factual/creative reference ($n = 30$)

E Full Head Sweep: Top 10 Heads

F Representation Analysis

Concept erasure. A logistic regression probe achieves 100% accuracy classifying self-referential vs. baseline from Layer 4 onward. Projecting out this classification direction and recomputing R_V: the effect is unchanged ($d = -1.82$ before, $d = -1.82$ after; $\Delta d = 0.005$). The geometric contraction operates in a *different subspace* from the linear probe's classification direction—R_V is not reducible to a single discriminant axis.

Distributed interchange intervention. At L5, individual PCA dimensions show $R_V \approx 1.0$ (near baseline); the effect accumulates only with the top-20 dimensions ($R_V = 2.184$). At L27, *every* individual PCA dimension shows $R_V \approx 0.41$ (strong contraction); top-20 combined: $R_V = 0.324$, $d = -3.42$. The late-layer contraction is a *global geometric property*, not a feature localized to specific rotated directions.

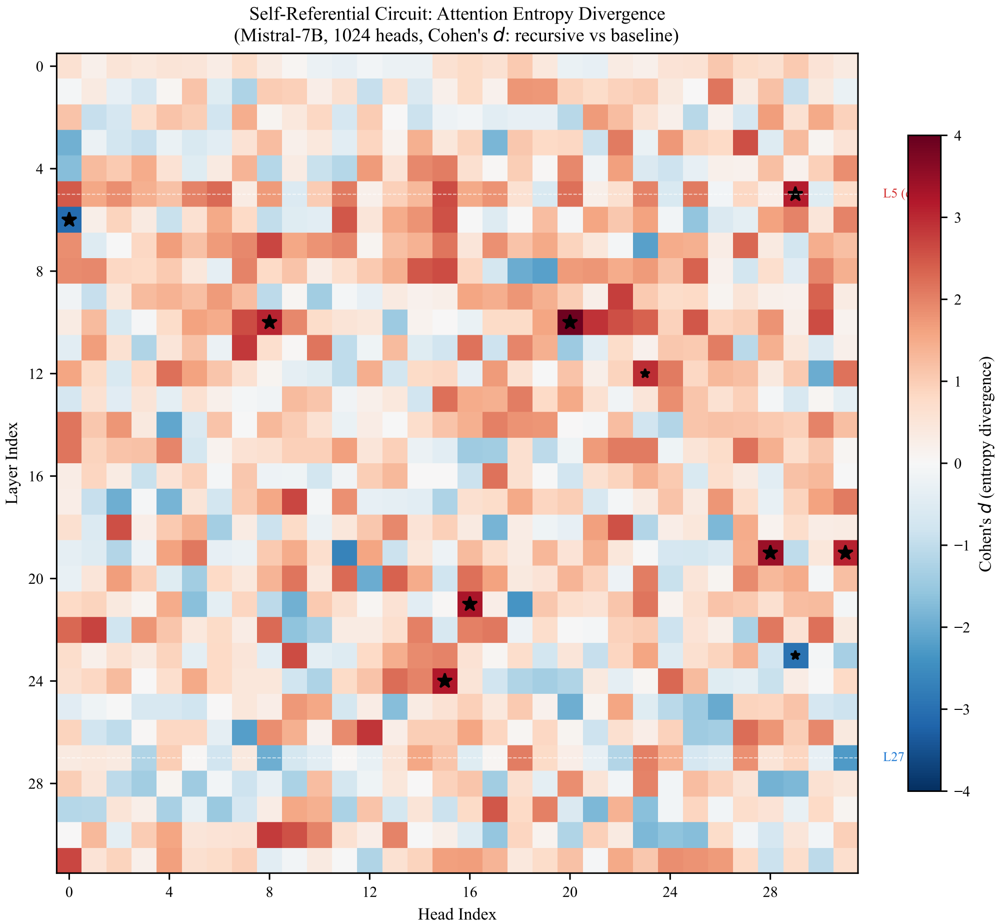


Figure 6: **Full 32×32 head sweep (Mistral-7B, $n = 100$ per condition).** Color encodes d for entropy separation between recursive and baseline prompts. 907 of 1,024 heads are significant on entropy ($p < 0.05$, uncorrected).

Table 8: Top 10 attention heads by entropy separation in the refreshed base $n = 100$ head sweep.

Layer	Head	d_{entropy}	Significance
20	09	+3.43	$p < 10^{-30}$
22	08	+3.37	$p < 10^{-30}$
24	14	+3.30	$p < 10^{-30}$
21	16	+3.17	$p < 10^{-29}$
21	22	+3.17	$p < 10^{-32}$
24	15	+3.11	$p < 10^{-30}$
26	16	+3.09	$p < 10^{-30}$
17	30	+3.09	$p < 10^{-29}$
23	27	+3.06	$p < 10^{-29}$
22	30	+3.06	$p < 10^{-32}$

Vocabulary projections. SVD vocabulary projections of the first singular direction in L27H10 show high scores for continuation/completion tokens and low scores for entity tokens, encoding a *continuation-vs-entity* gate. L27H31's second singular direction encodes a literal *self/other* axis: top tokens are "own", "Own", "answer" while bottom tokens are "same", "kind", "sorts".

G SVD Circuit Taxonomy

Table 9: SVD decomposition of seven circuit heads in Mistral-7B ($n = 100$ per condition).

Head	Role	d_{rank}	d_{top1}	Stability	SV1 Semantic Axis
L27H10	Compress	-1.32	1.36	0.94 / 0.54	exploratory token axis
L27H2	Compress	-1.35	1.51	0.90 / 0.79	exploratory token axis
L27H26	Compress	-0.95	0.45	0.20 / 0.34	exploratory token axis
L5H29	Diversify	$+0.94$	-0.51	0.52 / 0.42	exploratory token axis
L5H15	Neutral	-0.02	-0.14	0.60 / 0.43	exploratory token axis
L27H18	Compress	-0.74	0.84	0.70 / 0.30	exploratory token axis
L27H31	Neutral	-0.43	0.23	0.60 / 0.40	exploratory token axis

Role: Compress = rank contracts for recursive ($d < -0.5$); Diversify = rank expands ($d > 0.5$); Neutral = $|d| < 0.5$. Stability: mean cosine similarity of top-1 singular vector across prompts (recursive / baseline). The last column is a qualitative placeholder; the locked claims in this draft are the numeric rank and stability values.

H Theoretical Framework

H.1 Participation Ratio on the Grassmannian

The singular value decomposition $\mathbf{A}^{(l)} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ associates to each activation a point on the Grassmannian $\mathrm{Gr}(k, d)$ via the column space. The participation ratio is a spectral functional of the Gram matrix $\mathbf{G} = \mathbf{A}^\top\mathbf{A}$:

$$\mathrm{PR} = \frac{(\operatorname{tr}\mathbf{G})^2}{\operatorname{tr}\mathbf{G}^2} = \frac{\|\boldsymbol{\lambda}\|_1^2}{\|\boldsymbol{\lambda}\|_2^2} \tag{6}$$

where $\boldsymbol{\lambda}$ is the eigenvalue vector of $\mathbf{G}$.

H.2 Random Matrix Theory Baseline

Under the null hypothesis that activations are drawn i.i.d. from $\mathcal{N}(0, \sigma^2\mathbf{I})$, the eigenvalues follow the Marčenko-Pastur distribution with $\gamma = D/W$. For $W = 16$, $D = 4096$: $\mathrm{PR}_{\mathrm{MP}} \approx 15.94$ (near W). Observed recursive $\mathrm{PR}_{\mathrm{late}} \approx 5.8$ (Mistral-7B)—well below the random null, indicating structured spectral organization.

H.3 Information-Geometric Interpretation

Define geometric complexity as $C(l) = \log \mathrm{PR}(l)$. Then:

$$\log R_V = C(L_{\mathrm{late}}) - C(L_{\mathrm{early}}) \tag{7}$$

Negative $\log R_V$ means the network reduces geometric complexity across layers. Self-referential content produces the strongest reduction: representing "the system processing this text" requires fewer effective dimensions than representing external domains.